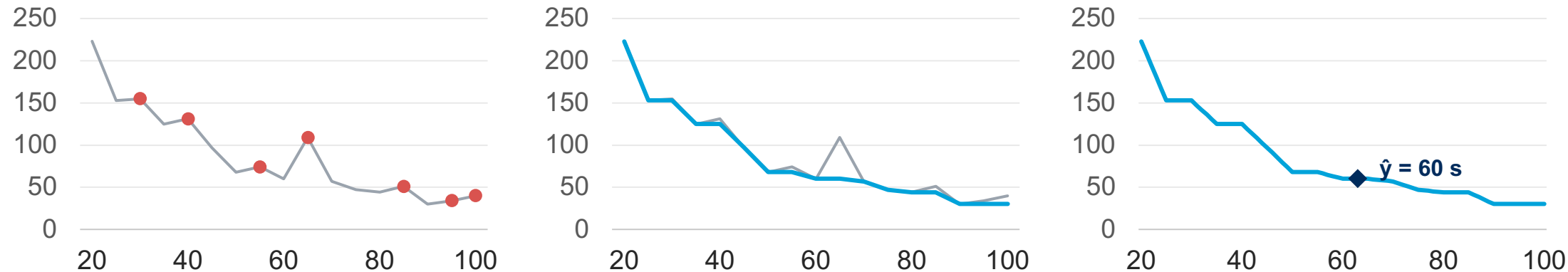
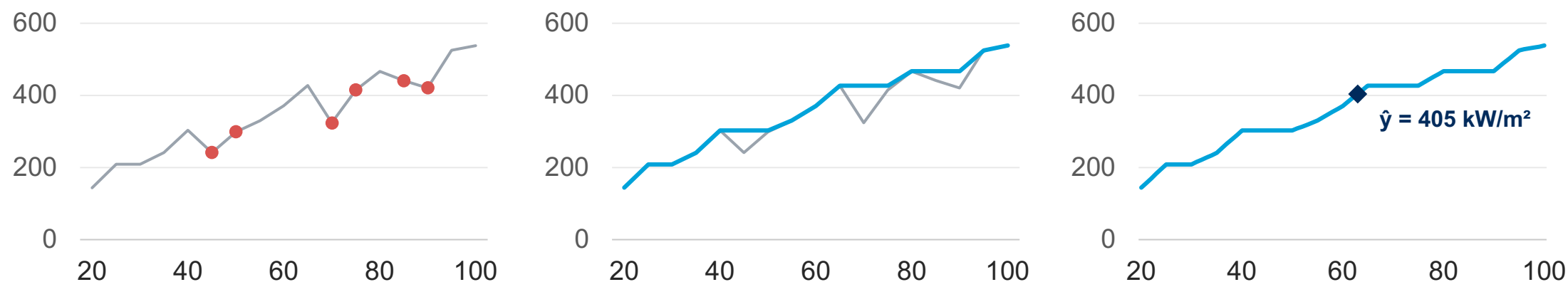
Monotone Spline Head

1 Raw MLP outputs at knots **2 Running min / max** **3 Interpolate at actual flux**

Time to ignition, t_{ig} (s): must be non-increasing in heat flux



Peak heat release rate, PHRR (kW/m²): must be non-decreasing in heat flux



x-axis: incident heat flux (kW/m²) ● raw output ● physics violation — monotone spline ◆ prediction at 63 kW/m²